

Credit Risk PD Modeling: Survival Analysis vs. Logistic Regression

Modeling time-to-default on ~2 million Lending Club loans using Kaplan-Meier estimation and Cox Proportional Hazards regression, benchmarked against logistic regression, and connected to IFRS 9 Expected Credit Loss (ECL) provisioning.

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Dataset: Lending Club Loan Data (Kaggle, public)

Tools: Python, pandas, lifelines, scikit-learn, matplotlib

Executive Summary

Most introductory credit risk projects predict whether a loan will default, using logistic regression on a binary outcome. This project instead models when a loan is likely to default, using survival analysis - Kaplan-Meier estimation and a Cox Proportional Hazards model - applied to roughly 2 million loans from Lending Club.

Three results stand out. First, loan grade is by far the strongest predictor of default risk: a grade G loan defaults at over five times the rate of a grade A loan, holding other factors constant. Second, the survival model outperforms a logistic regression fit on the same features (Concordance 0.69 vs. AUC 0.631), because it correctly distinguishes loans that have proven safe from loans that simply haven't had time to fail yet. Third, the survival curves translate directly into the inputs banks need for IFRS 9 provisioning - 12-month and lifetime probability of default - and reveal that the relative impact of moving a loan from a 12-month to a lifetime loss provision varies systematically by grade.

1. Data and Methodology

The analysis uses the public Lending Club loan dataset, filtered to loans with a definitive or ongoing status (Fully Paid, Current, Charged Off, or Default), totalling approximately 2.07 million loans after cleaning.

For each loan, two variables were constructed to enable survival analysis: a duration (the number of months between loan issuance and the last recorded payment) and an event indicator (1 if the loan was Charged Off or Default, 0 otherwise). Loans coded as 0 fall into two distinct groups - Fully Paid loans, which are known to have completed without defaulting, and Current loans, which are still active and simply have not defaulted yet. Both are treated as censored observations, a standard and necessary technique for handling outcomes that are only partially known.

Additional cleaning included converting loan term and employment length to numeric form (with a missingness flag for the roughly 39% of borrowers with no recorded employment length), and one-hot encoding categorical variables including loan grade, home ownership, loan purpose, and income verification status.

2. Kaplan-Meier Survival Analysis

2.1 Overall Survival Curve

The Kaplan-Meier estimator gives the probability that a loan has not yet defaulted at each point in its life. The overall curve declines steadily from 1.0 at issuance to approximately 0.69 by month 60 - meaning roughly 31% of loans default at some point over a typical loan term. Estimates beyond month 60 are based on a very small number of remaining loans and should not be relied upon.

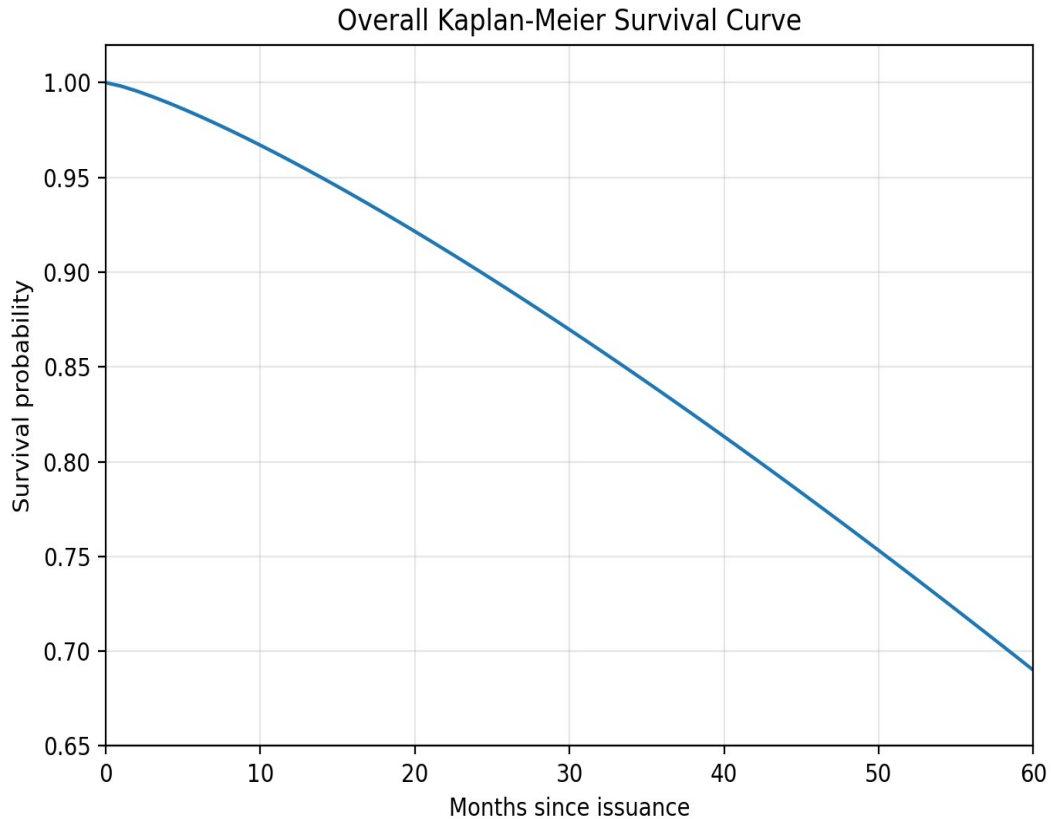


Figure 1. Overall Kaplan-Meier survival curve - probability of not having defaulted, by months since loan issuance.

2.2 Survival Curves by Loan Grade

Splitting the survival curve by Lending Club's assigned loan grade (A through G) produces a clean, monotonic separation. By month 60, grade A loans remain at roughly 0.89 survival (about 11% cumulative default), while grade G loans fall to roughly 0.37-0.40 (over 60% cumulative default). This is strong evidence that the grading system captures real, ordinal differences in default risk.

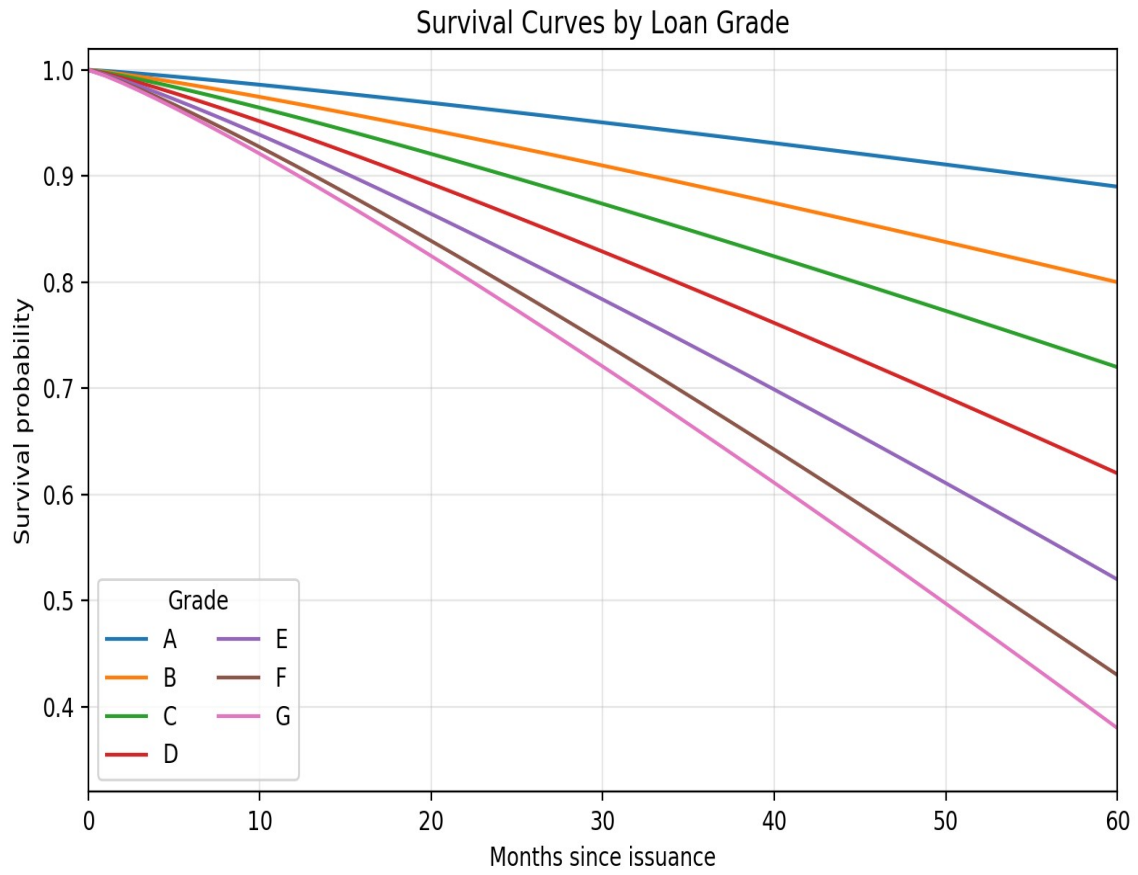


Figure 2. Kaplan-Meier survival curves stratified by loan grade (A-G).

3. Cox Proportional Hazards Model

A Cox Proportional Hazards model was fit on a 200,000-loan sample using 35 covariates, including loan terms, borrower financial characteristics, and the encoded categorical variables. The model achieved a Concordance of 0.69, meaning that when comparing two randomly chosen loans, the model correctly identifies which one defaulted first about 69% of the time.

Loan grade dominates the model. Figure 3 shows the hazard ratio for each grade relative to grade A (the reference category) - a hazard ratio of 5.22 for grade G means a grade G loan defaults at over five times the instantaneous rate of an otherwise identical grade A loan.

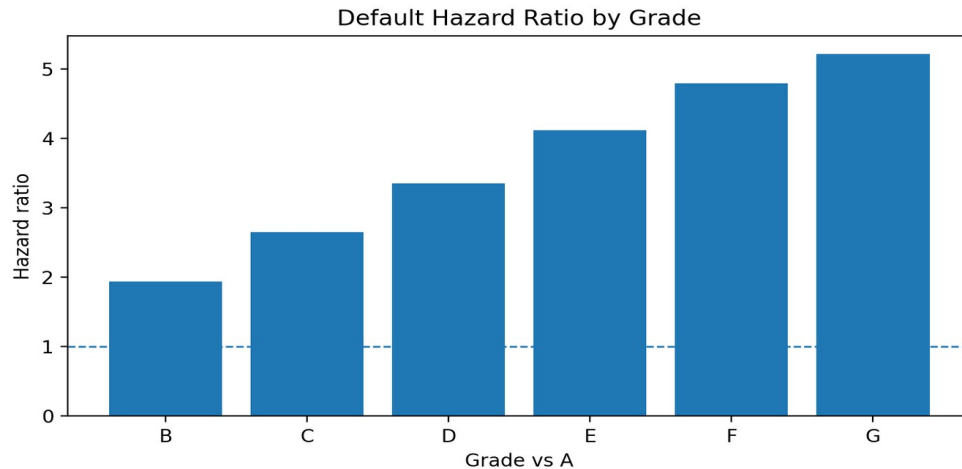


Figure 3. Default hazard ratio by loan grade, relative to grade A (Cox model).

Beyond grade, several other variables showed statistically significant effects, summarized below.

Variable	Hazard Ratio	Interpretation
Interest Rate	1.06	Each 1 percentage-point increase in interest rate raises the default hazard by roughly 6%.
Debt-to-Income (DTI)	~1.00	Higher DTI is associated with a small but statistically significant increase in default risk.
Annual Income	~1.00	Higher income is associated with slightly lower default risk; the per-dollar effect is small given the income scale.
Revolving Utilization	~1.00	Higher revolving credit utilization is associated with a small increase in default risk.
Verification: Verified	1.19	Verified loans show higher, not lower, default risk - likely because verification is applied selectively to riskier applications.
Home Ownership	~1.0-1.4	Not statistically significant ($p > 0.5$) once other factors are controlled for.

4. Model Comparison: Survival vs. Logistic Regression

A logistic regression was fit on the same 200,000-loan sample and the same 35 features, with the binary event indicator as the target. Its Area Under the ROC Curve (AUC) of 0.631 is noticeably lower than the Cox model's Concordance of 0.69, despite using identical inputs.

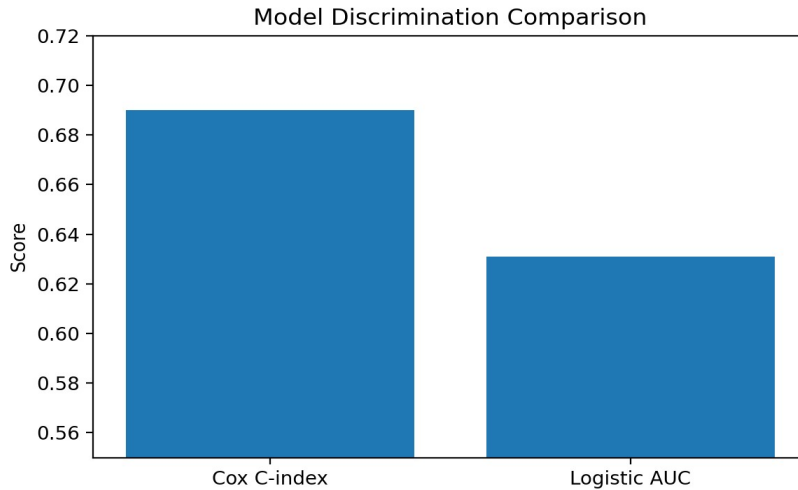


Figure 4. Discriminative power: Cox Proportional Hazards (Concordance) vs. Logistic Regression (AUC).

The gap is best explained by how each model treats loans that are still Current. Logistic regression assigns these loans the same label (0) as loans that have already completed their full term without defaulting - even though a loan only a few months old has simply not had time to reveal its outcome. The Cox model's censoring framework uses this partial information honestly, crediting each loan only with the time it has actually survived. This more accurate use of incomplete information is the core argument for modeling time-to-default rather than a simple binary outcome.

5. IFRS 9 / Expected Credit Loss Application

Under IFRS 9, loans are assigned to one of three stages, which determine how much loss provision a lender must hold. Stage 1 loans require a provision based on the probability of default over the next 12 months only. Stage 2 and 3 loans - those whose credit risk has increased significantly, or which show signs of impairment - require a provision based on the probability of default over the entire remaining life of the loan. This shift from a 12-month to a lifetime horizon, often called the 'staging cliff', can increase a loan's required provision several-fold.

The grade-level Kaplan-Meier curves give both quantities directly. Reading the survival probability at month 12 and month 36 for each grade, and converting to default probability (1 minus survival), produces the table below.

Grade	12-Month PD	36-Month PD	Cliff Ratio (36mo / 12mo)
A	1.4%	6.9%	4.92
B	3.5%	14.2%	4.07
C	6.4%	22.6%	3.51
D	10.1%	31.0%	3.08
E	14.4%	39.4%	2.75
F	19.7%	47.7%	2.41
G	25.8%	54.4%	2.11

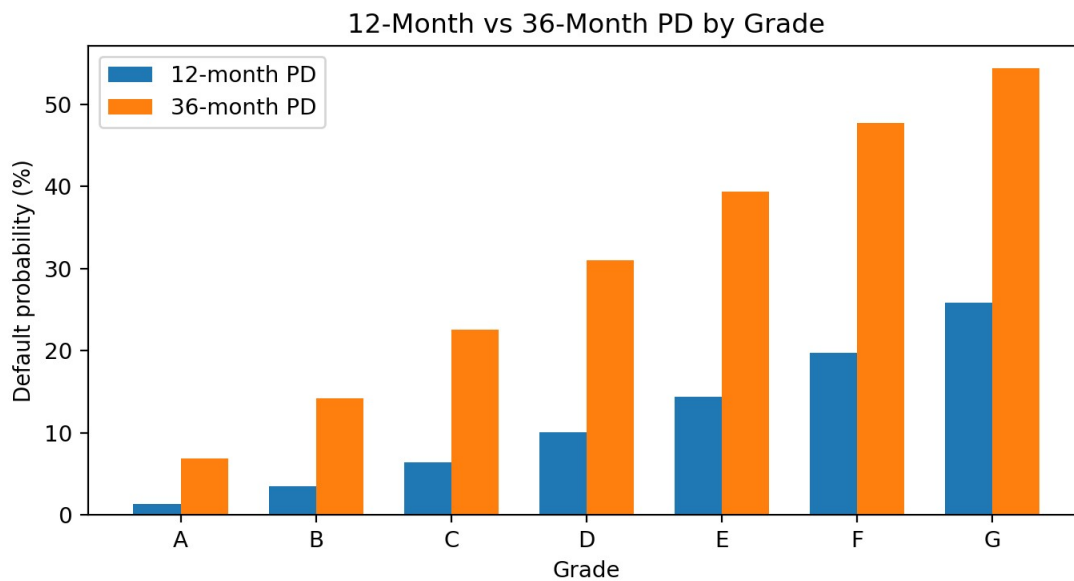


Figure 5. 12-month vs. 36-month default probability by loan grade.

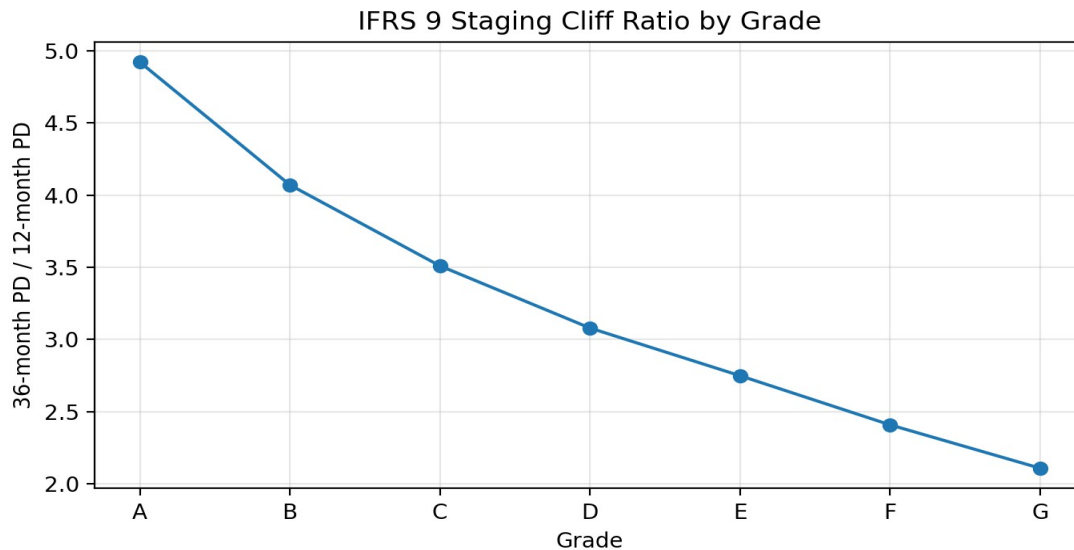


Figure 6. IFRS 9 staging cliff - lifetime PD as a multiple of 12-month PD, by grade.

Both PD measures rise steadily from grade A to grade G, as expected. The cliff ratio, however, moves in the opposite direction: it falls from 4.92 for grade A to 2.11 for grade G. For grade A, the 36-month default probability is nearly five times the 12-month figure, indicating that most of the (small) lifetime risk for high-grade loans is back-loaded - it tends to materialize later in the loan's life, often triggered by events unrelated to the original credit decision. For grade G, the 36-month figure is only about twice the 12-month figure, indicating that a much larger share of the (much larger) lifetime risk is front-loaded - financially stressed borrowers tend to default early if they default at all.

The practical implication is that a Stage 1-to-Stage 2 migration has a proportionally larger effect on required provisions for higher-grade loans than for lower-grade loans, even though the absolute provision for lower-grade loans remains far larger at every stage.

6. Limitations and Future Work

- The Cox model and logistic regression were fit on a 200,000-loan sample rather than the full ~2.07 million loans, for computational practicality.
- The proportional hazards assumption underlying the Cox model was not formally tested.
- Fully Paid and Current loans are both treated as censored. A competing-risks framework (e.g. a Fine-Gray model) would model payoff and default as distinct, competing outcomes.
- Only the Probability of Default (PD) component of Expected Credit Loss is modeled. Loss Given Default (LGD) and Exposure at Default (EAD) are out of scope.
- Tree-based survival models, such as Random Survival Forests, were considered as a non-linear benchmark but deprioritized in favor of interpretable models, consistent with how credit risk models are typically validated in regulated settings.

7. Conclusion

Framing credit risk as a time-to-default problem, rather than a binary classification problem, captures meaningfully more information from the same data - both in raw discriminative performance (0.69 vs. 0.631) and in how directly it maps onto real provisioning practice under IFRS 9. Loan grade remains the dominant driver of default risk, but the timing of that risk varies systematically across grades, with direct implications for how loss provisions evolve as a loan's credit quality changes.